

Questions

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1. Find the exact value of the improper integral

$$\int_0^4 \frac{1}{\sqrt{x(4-x)}} dx \quad [5]$$

Solution

Let

$$I = \int_0^4 \frac{1}{\sqrt{x(4-x)}} dx$$

This is an improper integral because the denominator is 0 at $x = 0$ and $x = 4$. So we write

$$I = \lim_{\substack{a \rightarrow 0^+ \\ b \rightarrow 4^-}} \int_a^b \frac{1}{\sqrt{x(4-x)}} dx$$

Now simplify the expression inside the square root:

$$x(4-x) = 4x - x^2 = 4 - (x^2 - 4x + 4) = 4 - (x-2)^2$$

So

$$I = \lim_{\substack{a \rightarrow 0^+ \\ b \rightarrow 4^-}} \int_a^b \frac{1}{\sqrt{4 - (x-2)^2}} dx$$

Using the standard result

$$\int \frac{1}{\sqrt{r^2 - x^2}} dx = \arcsin\left(\frac{x}{r}\right) + c$$

with $r = 2$ and x replaced by $x - 2$, we get

$$I = \lim_{\substack{a \rightarrow 0^+ \\ b \rightarrow 4^-}} \left[\arcsin\left(\frac{x-2}{2}\right) \right]_a^b$$

So

$$I = \lim_{\substack{a \rightarrow 0^+ \\ b \rightarrow 4^-}} \left(\arcsin\left(\frac{b-2}{2}\right) - \arcsin\left(\frac{a-2}{2}\right) \right)$$

Now

$$\frac{b-2}{2} \rightarrow 1 \quad \text{as } b \rightarrow 4^-$$

and

$$\frac{a-2}{2} \rightarrow -1 \quad \text{as } a \rightarrow 0^+$$

Therefore

$$I = \arcsin(1) - \arcsin(-1) = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

The exact value of the integral is π

2. Show that

$$\int_0^3 \frac{1}{x^2 - 2x + 5} dx = \frac{1}{2} \arctan(3) \quad [5]$$

Solution

Let

$$I = \int_0^3 \frac{1}{x^2 - 2x + 5} dx$$

First complete the square in the denominator:

$$x^2 - 2x + 5 = (x - 1)^2 + 4 = (x - 1)^2 + 2^2$$

So

$$I = \int_0^3 \frac{1}{(x - 1)^2 + 2^2} dx$$

Now use the standard result from the formula booklet:

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

Here $a = 2$, with x replaced by $x - 1$. Hence

$$I = \frac{1}{2} \left[\arctan\left(\frac{x - 1}{2}\right) \right]_0^3$$

Substituting the limits gives

$$\begin{aligned} I &= \frac{1}{2} \left(\arctan\left(\frac{3 - 1}{2}\right) - \arctan\left(\frac{0 - 1}{2}\right) \right) \\ &= \frac{1}{2} \left(\arctan(1) - \arctan\left(-\frac{1}{2}\right) \right) \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \arctan\left(\frac{1}{2}\right) \right) \end{aligned}$$

To simplify this, let

$$\theta = \frac{\pi}{4} + \arctan\left(\frac{1}{2}\right)$$

Then

$$\begin{aligned} \tan \theta &= \frac{\tan\left(\frac{\pi}{4}\right) + \tan\left(\arctan\left(\frac{1}{2}\right)\right)}{1 - \tan\left(\frac{\pi}{4}\right) \tan\left(\arctan\left(\frac{1}{2}\right)\right)} \\ &= \frac{1 + \frac{1}{2}}{1 - 1 \cdot \frac{1}{2}} \\ &= \frac{\frac{3}{2}}{\frac{1}{2}} \\ &= 3 \end{aligned}$$

Also, both $\frac{\pi}{4}$ and $\arctan\left(\frac{1}{2}\right)$ are between 0 and $\frac{\pi}{2}$, so θ is in the first quadrant. Therefore

$$\theta = \arctan(3)$$

Hence

$$I = \frac{1}{2} \arctan(3)$$

So

$$\int_0^3 \frac{1}{x^2 - 2x + 5} dx = \frac{1}{2} \arctan(3)$$

as required.

3. (i) Prove, from definitions involving exponentials, that

$$1 - \tanh^2 u = \operatorname{sech}^2 u \quad [3]$$

- (ii) Prove that, for $|y| < 1$,

$$\operatorname{artanh} y = \frac{1}{2} \ln \left(\frac{1+y}{1-y} \right) \quad [4]$$

- (iii) Use the substitution $x = 4 \tanh u$ to show that, for $|x| < 4$,

$$\int \frac{1}{16 - x^2} dx = \frac{1}{4} \operatorname{artanh} \frac{x}{4} + c$$

Hence deduce that

$$\int \frac{1}{16 - x^2} dx = \frac{1}{8} \ln \left(\frac{4+x}{4-x} \right) + c$$

where c is an arbitrary constant. [6]

- (iv) By first expressing $-t^2 + 6t + 7$ in completed square form, show that

$$\int_1^5 \frac{1}{-t^2 + 6t + 7} dt = \frac{1}{4} \ln 3 \quad [5]$$

Solution

- (i) Using the exponential definitions,

$$\tanh u = \frac{e^u - e^{-u}}{e^u + e^{-u}} \quad \text{and} \quad \operatorname{sech} u = \frac{2}{e^u + e^{-u}}$$

Then

$$\begin{aligned} 1 - \tanh^2 u &= 1 - \left(\frac{e^u - e^{-u}}{e^u + e^{-u}} \right)^2 \\ &= \frac{(e^u + e^{-u})^2 - (e^u - e^{-u})^2}{(e^u + e^{-u})^2} \\ &= \frac{(e^{2u} + 2 + e^{-2u}) - (e^{2u} - 2 + e^{-2u})}{(e^u + e^{-u})^2} \\ &= \frac{4}{(e^u + e^{-u})^2} \\ &= \left(\frac{2}{e^u + e^{-u}} \right)^2 \\ &= \operatorname{sech}^2 u \end{aligned}$$

Hence

$$1 - \tanh^2 u = \operatorname{sech}^2 u$$

- (ii) Let

$$u = \operatorname{artanh} y$$

so that

$$y = \tanh u$$

Using the exponential form of $\tanh u$,

$$y = \frac{e^u - e^{-u}}{e^u + e^{-u}}$$

Multiply top and bottom by e^u :

$$y = \frac{e^{2u} - 1}{e^{2u} + 1}$$

Now solve for u :

$$\begin{aligned} y(e^{2u} + 1) &= e^{2u} - 1 \\ ye^{2u} + y &= e^{2u} - 1 \\ e^{2u} - ye^{2u} &= 1 + y \\ e^{2u}(1 - y) &= 1 + y \\ e^{2u} &= \frac{1 + y}{1 - y} \end{aligned}$$

Since $|y| < 1$, both $1 + y$ and $1 - y$ are positive, so taking logarithms is valid:

$$2u = \ln \left(\frac{1 + y}{1 - y} \right)$$

Therefore

$$u = \frac{1}{2} \ln \left(\frac{1 + y}{1 - y} \right)$$

But $u = \operatorname{artanh} y$, so

$$\operatorname{artanh} y = \frac{1}{2} \ln \left(\frac{1 + y}{1 - y} \right)$$

(iii) Let

$$x = 4 \tanh u$$

Then

$$\frac{dx}{du} = 4 \operatorname{sech}^2 u \quad \Rightarrow \quad dx = 4 \operatorname{sech}^2 u \, du$$

Also,

$$\begin{aligned} 16 - x^2 &= 16 - 16 \tanh^2 u \\ &= 16(1 - \tanh^2 u) \\ &= 16 \operatorname{sech}^2 u \end{aligned}$$

using part (i).

So

$$\begin{aligned} \int \frac{1}{16 - x^2} dx &= \int \frac{1}{16 \operatorname{sech}^2 u} (4 \operatorname{sech}^2 u \, du) \\ &= \frac{1}{4} \int 1 \, du \\ &= \frac{u}{4} + c \end{aligned}$$

Now $x = 4 \tanh u$ gives

$$\tanh u = \frac{x}{4}$$

so

$$u = \operatorname{artanh} \frac{x}{4}$$

Hence

$$\int \frac{1}{16 - x^2} dx = \frac{1}{4} \operatorname{artanh} \frac{x}{4} + c$$

for $|x| < 4$.

Now use part (ii) with $y = \frac{x}{4}$. Since $|x| < 4$, we have $\left| \frac{x}{4} \right| < 1$, so

$$\operatorname{artanh} \frac{x}{4} = \frac{1}{2} \ln \left(\frac{1 + \frac{x}{4}}{1 - \frac{x}{4}} \right) = \frac{1}{2} \ln \left(\frac{4 + x}{4 - x} \right)$$

Therefore

$$\int \frac{1}{16-x^2} dx = \frac{1}{4} \cdot \frac{1}{2} \ln \left(\frac{4+x}{4-x} \right) + c = \frac{1}{8} \ln \left(\frac{4+x}{4-x} \right) + c$$

So

$$\int \frac{1}{16-x^2} dx = \frac{1}{8} \ln \left(\frac{4+x}{4-x} \right) + c$$

(iv) First complete the square:

$$\begin{aligned} -t^2 + 6t + 7 &= -(t^2 - 6t + 9) + 16 \\ &= 16 - (t-3)^2 \end{aligned}$$

So

$$\int_1^5 \frac{1}{-t^2 + 6t + 7} dt = \int_1^5 \frac{1}{16 - (t-3)^2} dt$$

Using part (iii), with x replaced by $t-3$,

$$\int \frac{1}{16 - (t-3)^2} dt = \frac{1}{8} \ln \left(\frac{4 + (t-3)}{4 - (t-3)} \right) + c$$

Therefore

$$\begin{aligned} \int_1^5 \frac{1}{-t^2 + 6t + 7} dt &= \left[\frac{1}{8} \ln \left(\frac{t+1}{7-t} \right) \right]_1^5 \\ &= \frac{1}{8} \ln \left(\frac{6}{2} \right) - \frac{1}{8} \ln \left(\frac{2}{6} \right) \\ &= \frac{1}{8} \ln 3 - \frac{1}{8} \ln \left(\frac{1}{3} \right) \\ &= \frac{1}{8} \ln 3 + \frac{1}{8} \ln 3 \\ &= \frac{1}{4} \ln 3 \end{aligned}$$

Hence

$$\int_1^5 \frac{1}{-t^2 + 6t + 7} dt = \frac{1}{4} \ln 3$$

4. (i) For $x > 5$, find

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}} dx$$

expressing your answer first in terms of arcosh and hence in logarithmic form.

[3]

- (ii) Use integration by substitution with $x - 3 = 2 \cosh u$ to show that, for $x > 5$,

$$\int \sqrt{x^2 - 6x + 5} dx = 2 \left(\frac{x-3}{2} \sqrt{\frac{(x-3)^2}{4} - 1} - \ln \left| \frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right| \right) + c$$

where c is an arbitrary constant.

[7]

Solution

- (i) First complete the square:

$$x^2 - 6x + 5 = (x - 3)^2 - 4$$

Since $x > 5$, we have $\frac{x-3}{2} > 1$, so we may let

$$x - 3 = 2 \cosh u \quad (u > 0)$$

Then

$$\frac{dx}{du} = 2 \sinh u \quad \Rightarrow \quad dx = 2 \sinh u du$$

and

$$\begin{aligned} \sqrt{x^2 - 6x + 5} &= \sqrt{(x-3)^2 - 4} \\ &= \sqrt{4 \cosh^2 u - 4} \\ &= 2 \sqrt{\cosh^2 u - 1} \\ &= 2 \sinh u \end{aligned}$$

since $u > 0$.

So

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 - 6x + 5}} dx &= \int \frac{1}{2 \sinh u} (2 \sinh u du) \\ &= \int 1 du \\ &= u + c \end{aligned}$$

Now

$$\cosh u = \frac{x-3}{2}$$

so

$$u = \operatorname{arcosh} \left(\frac{x-3}{2} \right)$$

Hence

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}} dx = \operatorname{arcosh} \left(\frac{x-3}{2} \right) + c$$

Using the standard result

$$\operatorname{arcosh} z = \ln \left(z + \sqrt{z^2 - 1} \right)$$

we get

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}} dx = \ln \left(\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right) + c$$

So the answer is

$$\operatorname{arcosh}\left(\frac{x-3}{2}\right) + c$$

and hence

$$\ln\left(\frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1}\right) + c$$

(ii) Let

$$I = \int \sqrt{x^2 - 6x + 5} \, dx$$

and use the substitution

$$x - 3 = 2 \cosh u \quad (x > 5)$$

Then, as above,

$$\sqrt{x^2 - 6x + 5} = 2 \sinh u \quad \text{and} \quad dx = 2 \sinh u \, du$$

So

$$\begin{aligned} I &= \int (2 \sinh u)(2 \sinh u) \, du \\ &= 4 \int \sinh^2 u \, du \end{aligned}$$

Use the identity

$$\cosh 2u = 2 \sinh^2 u + 1$$

so that

$$2 \sinh^2 u = \cosh 2u - 1$$

Therefore

$$\begin{aligned} I &= 2 \int (\cosh 2u - 1) \, du \\ &= 2 \left(\frac{1}{2} \sinh 2u - u \right) + c \\ &= \sinh 2u - 2u + c \end{aligned}$$

Now express this in terms of x .

Since

$$\cosh u = \frac{x-3}{2}$$

we have

$$\sinh u = \sqrt{\cosh^2 u - 1} = \sqrt{\frac{(x-3)^2}{4} - 1}$$

because $u > 0$.

Also

$$\begin{aligned} \sinh 2u &= 2 \sinh u \cosh u \\ &= 2 \left(\sqrt{\frac{(x-3)^2}{4} - 1} \right) \left(\frac{x-3}{2} \right) \\ &= (x-3) \sqrt{\frac{(x-3)^2}{4} - 1} \end{aligned}$$

Hence

$$\begin{aligned} I &= (x-3) \sqrt{\frac{(x-3)^2}{4} - 1} - 2u + c \\ &= 2 \left(\frac{x-3}{2} \sqrt{\frac{(x-3)^2}{4} - 1} - u \right) + c \end{aligned}$$

From part (ii),

$$u = \operatorname{arcosh} \left(\frac{x-3}{2} \right) = \ln \left| \frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right|$$

Therefore

$$\int \sqrt{x^2 - 6x + 5} \, dx = 2 \left(\frac{x-3}{2} \sqrt{\frac{(x-3)^2}{4} - 1} - \ln \left| \frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right| \right) + c$$

This is the required result.

5.

$$g(x) = \frac{2x^2 - 3x + 31}{x^3 - 3x^2 + 9x + 13}$$

(a) Express $g(x)$ in the form

$$\frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

where P and Q are constants to be found.

[3]

(b) Find $\int g(x) dx$, giving your answer in the form

$$A \ln |x+1| + B \arctan\left(\frac{x-2}{3}\right) + c$$

where A and B are constants to be found.

[4]

(c) Hence show that $\int_0^\infty g(x) dx$ diverges.

[2]

Solution

(a) First factorise the denominator:

$$x^3 - 3x^2 + 9x + 13 = (x+1)(x^2 - 4x + 13)$$

So

$$g(x) = \frac{2x^2 - 3x + 31}{(x+1)(x^2 - 4x + 13)}$$

and we write

$$\frac{2x^2 - 3x + 31}{(x+1)(x^2 - 4x + 13)} = \frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

Multiplying through by $(x+1)(x^2 - 4x + 13)$ gives

$$2x^2 - 3x + 31 = P(x^2 - 4x + 13) + Q(x+1)$$

Expand the right-hand side:

$$\begin{aligned} 2x^2 - 3x + 31 &= Px^2 - 4Px + 13P + Qx + Q \\ &= Px^2 + (-4P + Q)x + (13P + Q) \end{aligned}$$

Now compare coefficients with

$$2x^2 - 3x + 31$$

So

$$P = 2$$

and

$$-4P + Q = -3$$

Substituting $P = 2$:

$$\begin{aligned} -8 + Q &= -3 \\ Q &= 5 \end{aligned}$$

Checking the constant term:

$$13P + Q = 13(2) + 5 = 31$$

which is correct.

Hence

$$P = 2, \quad Q = 5$$

(b) From part (a),

$$g(x) = \frac{2}{x+1} + \frac{5}{x^2 - 4x + 13}$$

Complete the square in the quadratic:

$$x^2 - 4x + 13 = (x - 2)^2 + 9$$

So

$$g(x) = \frac{2}{x+1} + \frac{5}{(x-2)^2 + 9}$$

Therefore

$$\int g(x) \, dx = \int \frac{2}{x+1} \, dx + \int \frac{5}{(x-2)^2 + 9} \, dx$$

For the first integral,

$$\int \frac{2}{x+1} \, dx = 2 \ln |x+1|$$

For the second integral, use the standard result

$$\int \frac{1}{x^2 + a^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

With $a = 3$ and x replaced by $x - 2$,

$$\int \frac{5}{(x-2)^2 + 9} \, dx = 5 \int \frac{1}{(x-2)^2 + 3^2} \, dx = \frac{5}{3} \arctan\left(\frac{x-2}{3}\right)$$

Hence

$$\int g(x) \, dx = 2 \ln |x+1| + \frac{5}{3} \arctan\left(\frac{x-2}{3}\right) + c$$

So

$$A = 2, \quad B = \frac{5}{3}$$

(c) To consider the improper integral, write

$$\int_0^\infty g(x) \, dx = \lim_{t \rightarrow \infty} \int_0^t g(x) \, dx$$

Using the antiderivative from part (b),

$$\begin{aligned} \int_0^t g(x) \, dx &= \left[2 \ln |x+1| + \frac{5}{3} \arctan\left(\frac{x-2}{3}\right) \right]_0^t \\ &= 2 \ln(t+1) + \frac{5}{3} \arctan\left(\frac{t-2}{3}\right) - \left(2 \ln 1 + \frac{5}{3} \arctan\left(-\frac{2}{3}\right) \right) \\ &= 2 \ln(t+1) + \frac{5}{3} \arctan\left(\frac{t-2}{3}\right) + \frac{5}{3} \arctan\left(\frac{2}{3}\right) \end{aligned}$$

Now as $t \rightarrow \infty$,

$$\arctan\left(\frac{t-2}{3}\right) \rightarrow \frac{\pi}{2}$$

so that term stays finite, but

$$2 \ln(t+1) \rightarrow \infty$$

Therefore

$$\int_0^\infty g(x) \, dx$$

diverges, in fact it diverges to $+\infty$.

6. Show that

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}} dx = \frac{1}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) + \arcsin\left(\frac{1}{3}\right) \right) \quad [6]$$

Solution

Let

$$I = \int_1^4 \frac{1}{\sqrt{10+8x-2x^2}} dx$$

First complete the square in the expression under the root:

$$\begin{aligned} 10+8x-2x^2 &= -2(x^2-4x-5) \\ &= -2((x-2)^2-9) \\ &= 2(9-(x-2)^2) \end{aligned}$$

So

$$I = \int_1^4 \frac{1}{\sqrt{2(9-(x-2)^2)}} dx = \frac{1}{\sqrt{2}} \int_1^4 \frac{1}{\sqrt{9-(x-2)^2}} dx$$

Using the standard result

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right) + c$$

with $a = 3$ and x replaced by $x-2$,

$$\begin{aligned} I &= \frac{1}{\sqrt{2}} \left[\arcsin\left(\frac{x-2}{3}\right) \right]_1^4 \\ &= \frac{1}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) - \arcsin\left(-\frac{1}{3}\right) \right) \end{aligned}$$

Since $\arcsin(-a) = -\arcsin(a)$,

$$I = \frac{1}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) + \arcsin\left(\frac{1}{3}\right) \right)$$

Hence

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}} dx = \frac{1}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) + \arcsin\left(\frac{1}{3}\right) \right)$$

7. (a) Explain why

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx$$

is an improper integral.

[1]

(b) Show that

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx = k\pi$$

where k is a constant to be determined.

[4]

Solution

(a) The integral is improper because the interval of integration is unbounded: the upper limit is ∞ .

So it is defined as

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx = \lim_{b \rightarrow \infty} \int_0^b \frac{1}{x^2 - 6x + 18} dx$$

(b) Let

$$I = \int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx$$

First complete the square in the denominator:

$$x^2 - 6x + 18 = (x - 3)^2 + 9 = (x - 3)^2 + 3^2$$

Hence

$$I = \int_0^{\infty} \frac{1}{(x - 3)^2 + 3^2} dx$$

Since this is an improper integral, write it as a limit:

$$I = \lim_{b \rightarrow \infty} \int_0^b \frac{1}{(x - 3)^2 + 3^2} dx$$

Using the standard result

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 3$ and x replaced by $x - 3$, we get

$$\int \frac{1}{(x - 3)^2 + 3^2} dx = \frac{1}{3} \arctan\left(\frac{x - 3}{3}\right) + c$$

Therefore

$$\begin{aligned} I &= \lim_{b \rightarrow \infty} \left[\frac{1}{3} \arctan\left(\frac{x - 3}{3}\right) \right]_0^b \\ &= \frac{1}{3} \lim_{b \rightarrow \infty} \left(\arctan\left(\frac{b - 3}{3}\right) - \arctan(-1) \right) \end{aligned}$$

Now

$$\arctan\left(\frac{b - 3}{3}\right) \rightarrow \frac{\pi}{2} \quad \text{as } b \rightarrow \infty$$

and

$$\arctan(-1) = -\frac{\pi}{4}$$

So

$$\begin{aligned} I &= \frac{1}{3} \left(\frac{\pi}{2} - \left(-\frac{\pi}{4} \right) \right) \\ &= \frac{1}{3} \left(\frac{3\pi}{4} \right) \\ &= \frac{\pi}{4} \end{aligned}$$

Hence

$$\int_0^\infty \frac{1}{x^2 - 6x + 18} \, dx = \frac{\pi}{4} = k\pi$$

so

$$k = \frac{1}{4}$$

8. Evaluate

$$\int_0^4 \frac{1}{x^2 - 4x + 8} dx$$

giving your answer as an exact multiple of π .

[8]

Solution

First complete the square in the denominator:

$$x^2 - 4x + 8 = (x^2 - 4x + 4) + 4 = (x - 2)^2 + 4 = (x - 2)^2 + 2^2$$

So the integral becomes

$$\int_0^4 \frac{1}{x^2 - 4x + 8} dx = \int_0^4 \frac{1}{(x - 2)^2 + 2^2} dx$$

Now use the standard result from the formula booklet:

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 2$ and x replaced by $x - 2$.

Therefore

$$\int_0^4 \frac{1}{(x - 2)^2 + 2^2} dx = \left[\frac{1}{2} \arctan\left(\frac{x - 2}{2}\right) \right]_0^4$$

Now substitute the limits:

$$\begin{aligned} \left[\frac{1}{2} \arctan\left(\frac{x - 2}{2}\right) \right]_0^4 &= \frac{1}{2} \arctan\left(\frac{4 - 2}{2}\right) - \frac{1}{2} \arctan\left(\frac{0 - 2}{2}\right) \\ &= \frac{1}{2} \arctan(1) - \frac{1}{2} \arctan(-1) \\ &= \frac{1}{2} \cdot \frac{\pi}{4} - \frac{1}{2} \cdot \left(-\frac{\pi}{4}\right) \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{\pi}{4}\right) \\ &= \frac{1}{2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{4} \end{aligned}$$

Hence

$$\int_0^4 \frac{1}{x^2 - 4x + 8} dx = \frac{\pi}{4}$$

So the exact value, as a multiple of π , is $\frac{\pi}{4}$.

9.

$$f(x) = \frac{6x+1}{4x^2-8x+13}$$

(a) Find $\int f(x) dx$, giving your answer in the form

$$A \ln(4x^2 - 8x + 13) + B \arctan\left(\frac{2x-2}{3}\right) + c$$

where c is an arbitrary constant and A and B are constants to be found.

[4]

(b) Hence show that the mean value of $f(x)$ over the interval $[1, \frac{5}{2}]$ is

$$\frac{1}{2} \ln 2 + \frac{7\pi}{36}$$

[3]

(c) Hence write down the mean value of $3 - 2f(x)$ over the interval $[1, \frac{5}{2}]$.

[1]

Solution

(a) Let

$$I = \int \frac{6x+1}{4x^2-8x+13} dx$$

Write the numerator in terms of the derivative of the denominator:

$$6x+1 = \frac{3}{4}(8x-8) + 7$$

So

$$I = \frac{3}{4} \int \frac{8x-8}{4x^2-8x+13} dx + 7 \int \frac{1}{4x^2-8x+13} dx$$

For the first integral,

$$\int \frac{g'(x)}{g(x)} dx = \ln |g(x)| + c$$

and here

$$4x^2 - 8x + 13 = (2x-2)^2 + 9 > 0$$

so no modulus is needed. Hence

$$\frac{3}{4} \int \frac{8x-8}{4x^2-8x+13} dx = \frac{3}{4} \ln(4x^2-8x+13)$$

For the second integral, complete the square:

$$4x^2 - 8x + 13 = (2x-2)^2 + 9$$

Using the standard result

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 3$ and x replaced by $2x-2$, noting that $\frac{d}{dx}(2x-2) = 2$,

$$\begin{aligned} 7 \int \frac{1}{(2x-2)^2+9} dx &= 7 \cdot \frac{1}{2} \cdot \frac{1}{3} \arctan\left(\frac{2x-2}{3}\right) \\ &= \frac{7}{6} \arctan\left(\frac{2x-2}{3}\right) \end{aligned}$$

Therefore

$$\int f(x) \, dx = \frac{3}{4} \ln(4x^2 - 8x + 13) + \frac{7}{6} \arctan\left(\frac{2x-2}{3}\right) + c$$

Therefore

$$A = \frac{3}{4}, \quad B = \frac{7}{6}$$

(b) From part (a), an antiderivative is

$$F(x) = \frac{3}{4} \ln(4x^2 - 8x + 13) + \frac{7}{6} \arctan\left(\frac{2x-2}{3}\right)$$

The mean value of $f(x)$ over $\left[1, \frac{5}{2}\right]$ is

$$\frac{1}{\frac{5}{2} - 1} \int_1^{5/2} f(x) \, dx = \frac{2}{3} (F(5/2) - F(1))$$

Now

$$\begin{aligned} F\left(\frac{5}{2}\right) &= \frac{3}{4} \ln\left(4 \cdot \frac{25}{4} - 8 \cdot \frac{5}{2} + 13\right) + \frac{7}{6} \arctan\left(\frac{2 \cdot \frac{5}{2} - 2}{3}\right) \\ &= \frac{3}{4} \ln(18) + \frac{7}{6} \arctan(1) \\ &= \frac{3}{4} \ln(18) + \frac{7\pi}{24} \end{aligned}$$

Also

$$\begin{aligned} F(1) &= \frac{3}{4} \ln(4 - 8 + 13) + \frac{7}{6} \arctan(0) \\ &= \frac{3}{4} \ln(9) \end{aligned}$$

So

$$\begin{aligned} \frac{2}{3} (F(5/2) - F(1)) &= \frac{2}{3} \left(\frac{3}{4} \ln \frac{18}{9} + \frac{7\pi}{24} \right) \\ &= \frac{2}{3} \left(\frac{3}{4} \ln 2 + \frac{7\pi}{24} \right) \\ &= \frac{1}{2} \ln 2 + \frac{7\pi}{36} \end{aligned}$$

Hence the mean value is

$$\frac{1}{2} \ln 2 + \frac{7\pi}{36}$$

(c) Using linearity of the mean value,

$$\text{mean value of } (3 - 2f(x)) = 3 - 2 \times \text{mean value of } f(x)$$

So

$$\begin{aligned} \text{mean value of } (3 - 2f(x)) &= 3 - 2 \left(\frac{1}{2} \ln 2 + \frac{7\pi}{36} \right) \\ &= 3 - \ln 2 - \frac{7\pi}{18} \end{aligned}$$

Therefore the mean value is

$$3 - \ln 2 - \frac{7\pi}{18}$$

10.

$$g(x) = \frac{4x - 1}{x^2 - 4x + 8}$$

Show that

$$\int g(x) \, dx = A \arctan\left(\frac{x-2}{2}\right) + B \ln(x^2 - 4x + 8) + c$$

where c is an arbitrary constant and A and B are constants to be found.

[5]

Solution

First complete the square in the denominator:

$$x^2 - 4x + 8 = (x - 2)^2 + 4$$

Now rewrite the numerator so that part of it is the derivative of the denominator:

$$4x - 1 = 2(2x - 4) + 7$$

So

$$\begin{aligned} g(x) &= \frac{4x - 1}{x^2 - 4x + 8} \\ &= \frac{2(2x - 4) + 7}{x^2 - 4x + 8} \\ &= 2 \frac{2x - 4}{x^2 - 4x + 8} + \frac{7}{(x - 2)^2 + 4} \end{aligned}$$

Hence

$$\int g(x) \, dx = 2 \int \frac{2x - 4}{x^2 - 4x + 8} \, dx + 7 \int \frac{1}{(x - 2)^2 + 4} \, dx$$

For the first integral, use

$$\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)| + c$$

with $f(x) = x^2 - 4x + 8$. Then

$$2 \int \frac{2x - 4}{x^2 - 4x + 8} \, dx = 2 \ln(x^2 - 4x + 8)$$

since $x^2 - 4x + 8 = (x - 2)^2 + 4 > 0$.

For the second integral, use the standard result

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 2$ and x replaced by $x - 2$. Hence

$$7 \int \frac{1}{(x - 2)^2 + 4} \, dx = 7 \cdot \frac{1}{2} \arctan\left(\frac{x - 2}{2}\right)$$

Therefore

$$\int g(x) \, dx = 2 \ln(x^2 - 4x + 8) + \frac{7}{2} \arctan\left(\frac{x - 2}{2}\right) + c$$

This is of the form

$$\int g(x) \, dx = A \arctan\left(\frac{x - 2}{2}\right) + B \ln(x^2 - 4x + 8) + c$$

with

$$A = \frac{7}{2}, \quad B = 2$$

So the required constants are $A = \frac{7}{2}$ and $B = 2$.

11. You are given that $y = \operatorname{arcosh} x$, where $x \geq 1$.

(i) Show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}} \quad [4]$$

(ii) Show, by integrating the result in part (i), that

$$y = \ln \left(x + \sqrt{x^2 - 1} \right) \quad [4]$$

(iii) Hence show that

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2 - 9}} dx = \frac{1}{2} \left(\operatorname{arcosh} \frac{5}{3} - \operatorname{arcosh} \frac{5}{4} \right)$$

Express this answer in logarithmic form.

[4]

Solution

(i) Since $y = \operatorname{arcosh} x$, we have

$$\cosh y = x$$

Differentiate implicitly with respect to x :

$$\sinh y \frac{dy}{dx} = 1$$

So

$$\frac{dy}{dx} = \frac{1}{\sinh y}$$

Using the identity

$$\cosh^2 y - \sinh^2 y = 1$$

gives

$$\sinh^2 y = \cosh^2 y - 1 = x^2 - 1$$

Also $y = \operatorname{arcosh} x$ with $x \geq 1$, so $y \geq 0$, hence $\sinh y \geq 0$. Therefore

$$\sinh y = \sqrt{x^2 - 1}$$

and so

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$$

as required.

(ii) Integrating the result from part (i),

$$y = \int \frac{1}{\sqrt{x^2 - 1}} dx$$

Use the standard result

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

With $a = 1$,

$$y = \ln \left| x + \sqrt{x^2 - 1} \right| + C$$

Since $x \geq 1$, the quantity $x + \sqrt{x^2 - 1}$ is positive, so

$$y = \ln \left(x + \sqrt{x^2 - 1} \right) + C$$

Now when $x = 1$,

$$y = \operatorname{arcosh} 1 = 0$$

Hence

$$0 = \ln(1 + \sqrt{1^2 - 1}) + C = \ln 1 + C = C$$

So $C = 0$, and therefore

$$y = \ln(x + \sqrt{x^2 - 1})$$

Alternative method (without quoting the standard result):

Let

$$I = \int \frac{1}{\sqrt{x^2 - 1}} dx$$

Multiply the integrand by

$$\frac{x + \sqrt{x^2 - 1}}{x + \sqrt{x^2 - 1}}$$

to obtain

$$\begin{aligned} I &= \int \frac{x + \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}(x + \sqrt{x^2 - 1})} dx \\ &= \int \frac{1 + \frac{x}{\sqrt{x^2 - 1}}}{x + \sqrt{x^2 - 1}} dx \end{aligned}$$

Now

$$\frac{d}{dx}(x + \sqrt{x^2 - 1}) = 1 + \frac{x}{\sqrt{x^2 - 1}}$$

so

$$I = \int \frac{f'(x)}{f(x)} dx \quad \text{with} \quad f(x) = x + \sqrt{x^2 - 1}$$

Hence

$$I = \ln(x + \sqrt{x^2 - 1}) + C$$

since $x \geq 1$ implies $x + \sqrt{x^2 - 1} > 0$

Therefore

$$y = \ln(x + \sqrt{x^2 - 1}) + C$$

Now when $x = 1$,

$$y = \operatorname{arcosh} 1 = 0$$

so

$$0 = \ln(1 + \sqrt{1^2 - 1}) + C = \ln 1 + C = C$$

Thus $C = 0$, giving

$$y = \ln(x + \sqrt{x^2 - 1})$$

(iii) Since

$$\sqrt{4x^2 - 9} = 3\sqrt{\left(\frac{2x}{3}\right)^2 - 1}$$

and

$$\frac{d}{dx}\left(\frac{2x}{3}\right) = \frac{2}{3}$$

we use the result from part (i), with x replaced by $\frac{2x}{3}$:

$$\int \frac{1}{\sqrt{4x^2 - 9}} dx = \frac{1}{2} \operatorname{arcosh}\left(\frac{2x}{3}\right) + c$$

Therefore

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2-9}} dx = \left[\frac{1}{2} \operatorname{arcosh} \left(\frac{2x}{3} \right) \right]_{15/8}^{5/2} = \frac{1}{2} \left(\operatorname{arcosh} \frac{5}{3} - \operatorname{arcosh} \frac{5}{4} \right)$$

which is the required result.

Now express this in logarithmic form using part (iii):

$$\operatorname{arcosh} t = \ln \left(t + \sqrt{t^2 - 1} \right)$$

Hence

$$\operatorname{arcosh} \frac{5}{3} = \ln \left(\frac{5}{3} + \sqrt{\frac{25}{9} - 1} \right) = \ln \left(\frac{5}{3} + \frac{4}{3} \right) = \ln 3$$

and

$$\operatorname{arcosh} \frac{5}{4} = \ln \left(\frac{5}{4} + \sqrt{\frac{25}{16} - 1} \right) = \ln \left(\frac{5}{4} + \frac{3}{4} \right) = \ln 2$$

Therefore

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2-9}} dx = \frac{1}{2} (\ln 3 - \ln 2) = \frac{1}{2} \ln \left(\frac{3}{2} \right)$$

So the logarithmic form is

$$\frac{1}{2} \ln \left(\frac{3}{2} \right)$$

12. Determine each of the following integrals.

(i)

$$\int \frac{1}{9x^2 + 12x + 20} dx \quad [4]$$

(ii)

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}} dx \quad [4]$$

Solution

(i) Complete the square in the denominator:

$$\begin{aligned} 9x^2 + 12x + 20 &= 9 \left(x^2 + \frac{4}{3}x \right) + 20 \\ &= 9 \left[\left(x + \frac{2}{3} \right)^2 - \frac{4}{9} \right] + 20 \\ &= 9 \left(x + \frac{2}{3} \right)^2 - 4 + 20 \\ &= 9 \left(x + \frac{2}{3} \right)^2 + 16 \\ &= 9 \left[\left(x + \frac{2}{3} \right)^2 + \left(\frac{4}{3} \right)^2 \right] \end{aligned}$$

So

$$\int \frac{1}{9x^2 + 12x + 20} dx = \frac{1}{9} \int \frac{1}{\left(x + \frac{2}{3} \right)^2 + \left(\frac{4}{3} \right)^2} dx$$

Using the standard result

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

with $a = \frac{4}{3}$ and x replaced by $x + \frac{2}{3}$,

$$\begin{aligned} \int \frac{1}{9x^2 + 12x + 20} dx &= \frac{1}{9} \cdot \frac{1}{4/3} \arctan\left(\frac{x + \frac{2}{3}}{4/3}\right) + C \\ &= \frac{1}{12} \arctan\left(\frac{3x + 2}{4}\right) + C \end{aligned}$$

Hence,

$$\int \frac{1}{9x^2 + 12x + 20} dx = \frac{1}{12} \arctan\left(\frac{3x + 2}{4}\right) + C$$

(ii) Rewrite the expression under the root in completed-square form:

$$\begin{aligned}45 + 12x - 9x^2 &= -9 \left(x^2 - \frac{4}{3}x - 5 \right) \\&= -9 \left[\left(x - \frac{2}{3} \right)^2 - \frac{4}{9} - 5 \right] \\&= -9 \left[\left(x - \frac{2}{3} \right)^2 - \frac{49}{9} \right] \\&= 9 \left[\frac{49}{9} - \left(x - \frac{2}{3} \right)^2 \right] \\&= 9 \left[\left(\frac{7}{3} \right)^2 - \left(x - \frac{2}{3} \right)^2 \right]\end{aligned}$$

So

$$\begin{aligned}\int \frac{1}{\sqrt{45 + 12x - 9x^2}} dx &= \int \frac{1}{\sqrt{9 \left[\left(\frac{7}{3} \right)^2 - \left(x - \frac{2}{3} \right)^2 \right]}} dx \\&= \frac{1}{3} \int \frac{1}{\sqrt{\left(\frac{7}{3} \right)^2 - \left(x - \frac{2}{3} \right)^2}} dx\end{aligned}$$

Using the standard result

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C$$

with $a = \frac{7}{3}$ and x replaced by $x - \frac{2}{3}$,

$$\begin{aligned}\int \frac{1}{\sqrt{45 + 12x - 9x^2}} dx &= \frac{1}{3} \arcsin\left(\frac{x - \frac{2}{3}}{7/3}\right) + C \\&= \frac{1}{3} \arcsin\left(\frac{3x - 2}{7}\right) + C\end{aligned}$$

Hence,

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}} dx = \frac{1}{3} \arcsin\left(\frac{3x - 2}{7}\right) + C$$

13. Without using a calculator, find

(a)

$$\int_{-1/5}^1 \frac{1}{25x^2 - 20x + 13} dx$$

giving your answer as a multiple of π

[5]

(b)

$$\int_{-1/5}^1 \frac{1}{\sqrt{25x^2 - 20x + 13}} dx$$

giving your answer in the form $p \ln(q + r\sqrt{2})$, where p , q and r are rational numbers to be found.

[7]

Solution

(a) Complete the square in the denominator:

$$25x^2 - 20x + 13 = (5x - 2)^2 + 9$$

So

$$\int_{-1/5}^1 \frac{1}{25x^2 - 20x + 13} dx = \int_{-1/5}^1 \frac{1}{(5x - 2)^2 + 9} dx$$

Using the standard result

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 3$ and x replaced by $5x - 2$, noting that $\frac{d}{dx}(5x - 2) = 5$, we get

$$\begin{aligned} \int_{-1/5}^1 \frac{1}{(5x - 2)^2 + 9} dx &= \left[\frac{1}{15} \arctan\left(\frac{5x - 2}{3}\right) \right]_{-1/5}^1 \\ &= \frac{1}{15} (\arctan 1 - \arctan(-1)) \\ &= \frac{1}{15} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right) \\ &= \frac{1}{15} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{30} \end{aligned}$$

Therefore, the value of the integral is

$$\frac{\pi}{30}$$

(b) As in part (a),

$$25x^2 - 20x + 13 = (5x - 2)^2 + 9$$

So

$$\int_{-1/5}^1 \frac{1}{\sqrt{25x^2 - 20x + 13}} dx = \int_{-1/5}^1 \frac{1}{\sqrt{(5x - 2)^2 + 9}} dx$$

Using the standard result

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln|x + \sqrt{x^2 + a^2}| + c$$

with $a = 3$ and x replaced by $5x - 2$, noting that $\frac{d}{dx}(5x - 2) = 5$, we have

$$\int \frac{1}{\sqrt{(5x-2)^2+9}} dx = \frac{1}{5} \ln \left| (5x-2) + \sqrt{(5x-2)^2+9} \right| + c$$

Hence

$$\begin{aligned} \int_{-\frac{1}{5}}^1 \frac{1}{\sqrt{(5x-2)^2+9}} dx &= \left[\frac{1}{5} \ln \left| (5x-2) + \sqrt{(5x-2)^2+9} \right| \right]_{-\frac{1}{5}}^1 \\ &= \frac{1}{5} \left(\ln(3+3\sqrt{2}) - \ln(-3+3\sqrt{2}) \right) \\ &= \frac{1}{5} \ln \left(\frac{1+\sqrt{2}}{\sqrt{2}-1} \right) \end{aligned}$$

Now simplify the fraction by rationalising:

$$\begin{aligned} \frac{1+\sqrt{2}}{\sqrt{2}-1} &= \frac{(1+\sqrt{2})(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)} \\ &= \frac{(1+\sqrt{2})^2}{2-1} \\ &= (1+\sqrt{2})^2 \\ &= 1+2\sqrt{2}+2 \\ &= 3+2\sqrt{2} \end{aligned}$$

So the integral is

$$\frac{1}{5} \ln(3+2\sqrt{2})$$

Therefore, in the form $p \ln(q+r\sqrt{2})$,

$$p = \frac{1}{5}, \quad q = 3, \quad r = 2$$

14. (i) Given that $\tanh y = x$, where $|x| < 1$, show that

$$y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

and hence that

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \quad [6]$$

- (ii) Use your result in part (i), or otherwise, to find

$$\int_{1/5}^{2/5} \frac{1}{4-9x^2} dx$$

expressing your answer in an exact logarithmic form.

[6]

Solution

- (i) Starting from the definition of the hyperbolic tangent,

$$\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

Multiplying numerator and denominator by e^y gives

$$\tanh y = \frac{e^{2y} - 1}{e^{2y} + 1}$$

Given that $\tanh y = x$, we have

$$\begin{aligned} x &= \frac{e^{2y} - 1}{e^{2y} + 1} \\ x(e^{2y} + 1) &= e^{2y} - 1 \\ xe^{2y} + x &= e^{2y} - 1 \\ e^{2y} - xe^{2y} &= x + 1 \\ e^{2y}(1 - x) &= 1 + x \end{aligned}$$

Since $|x| < 1$, we have $1 - x > 0$, so we may divide through by $1 - x$:

$$e^{2y} = \frac{1+x}{1-x}$$

Now take natural logarithms:

$$2y = \ln \left(\frac{1+x}{1-x} \right)$$

Hence

$$y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

So the required result is shown.

Since $\operatorname{artanh} x$ is the inverse of $\tanh y$, this means

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

- (ii) From part (i),

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

for $|x| < 1$.

Differentiating,

$$\begin{aligned}\frac{d}{dx}(\operatorname{artanh} x) &= \frac{1}{2} \frac{d}{dx} [\ln(1+x) - \ln(1-x)] \\ &= \frac{1}{2} \left(\frac{1}{1+x} + \frac{1}{1-x} \right) \\ &= \frac{1}{2} \left(\frac{(1-x) + (1+x)}{1-x^2} \right) \\ &= \frac{1}{1-x^2}\end{aligned}$$

So

$$\int \frac{1}{1-x^2} dx = \operatorname{artanh} x + c$$

Now

$$4 - 9x^2 = 4 \left(1 - \left(\frac{3x}{2} \right)^2 \right)$$

and

$$\frac{d}{dx} \left(\frac{3x}{2} \right) = \frac{3}{2}$$

Therefore

$$\int_{1/5}^{2/5} \frac{1}{4-9x^2} dx = \frac{1}{6} \left[\operatorname{artanh} \left(\frac{3x}{2} \right) \right]_{1/5}^{2/5}$$

Using the result from part (i),

$$\begin{aligned}\int_{1/5}^{2/5} \frac{1}{4-9x^2} dx &= \frac{1}{6} \left[\frac{1}{2} \ln \left(\frac{1 + \frac{3x}{2}}{1 - \frac{3x}{2}} \right) \right]_{1/5}^{2/5} \\ &= \frac{1}{12} \left[\ln \left(\frac{1 + \frac{3x}{2}}{1 - \frac{3x}{2}} \right) \right]_{1/5}^{2/5} \\ &= \frac{1}{12} \left(\ln \left(\frac{1 + \frac{3}{5}}{1 - \frac{3}{5}} \right) - \ln \left(\frac{1 + \frac{3}{10}}{1 - \frac{3}{10}} \right) \right) \\ &= \frac{1}{12} \left(\ln 4 - \ln \left(\frac{13}{7} \right) \right) \\ &= \frac{1}{12} \ln \left(\frac{28}{13} \right)\end{aligned}$$

Hence the exact value of the integral is

$$\frac{1}{12} \ln \left(\frac{28}{13} \right)$$

15. Show that

$$\int \frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} dx = A \ln|x-1| + B \ln(x^2 + 4x + 13) + D \arctan\left(\frac{x+2}{3}\right) + c$$

where A , B and D are constants to be found.

[7]

Solution

Write

$$\frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} = \frac{P}{x-1} + \frac{Qx + R}{x^2 + 4x + 13}$$

Then

$$x^2 + 14x + 21 = P(x^2 + 4x + 13) + (Qx + R)(x-1)$$

Expanding the right-hand side,

$$\begin{aligned} P(x^2 + 4x + 13) + (Qx + R)(x-1) &= Px^2 + 4Px + 13P + Qx^2 - Qx + Rx - R \\ &= (P+Q)x^2 + (4P-Q+R)x + (13P-R) \end{aligned}$$

Comparing coefficients with $x^2 + 14x + 21$ gives

$$\begin{aligned} P + Q &= 1 \\ 4P - Q + R &= 14 \\ 13P - R &= 21 \end{aligned}$$

From the first equation,

$$Q = 1 - P$$

From the third equation,

$$R = 13P - 21$$

Substitute into the second equation:

$$\begin{aligned} 4P - (1 - P) + (13P - 21) &= 14 \\ 4P - 1 + P + 13P - 21 &= 14 \\ 18P - 22 &= 14 \\ 18P &= 36 \\ P &= 2 \end{aligned}$$

So

$$Q = 1 - 2 = -1, \quad R = 13(2) - 21 = 5$$

Hence

$$\frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} = \frac{2}{x-1} + \frac{-x+5}{x^2 + 4x + 13}$$

Therefore

$$\int \frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} dx = \int \frac{2}{x-1} dx + \int \frac{-x+5}{x^2 + 4x + 13} dx$$

Now rewrite the second numerator in terms of the derivative of the quadratic:

$$-x + 5 = -\frac{1}{2}(2x + 4) + 7$$

Also,

$$x^2 + 4x + 13 = (x+2)^2 + 9$$

So

$$\begin{aligned} \int \frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} dx &= \int \frac{2}{x-1} dx - \frac{1}{2} \int \frac{2x+4}{x^2 + 4x + 13} dx + 7 \int \frac{1}{(x+2)^2 + 9} dx \\ &= 2 \ln|x-1| - \frac{1}{2} \ln(x^2 + 4x + 13) + 7 \int \frac{1}{(x+2)^2 + 3^2} dx \end{aligned}$$

Using the standard result

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 3$ and x replaced by $x + 2$,

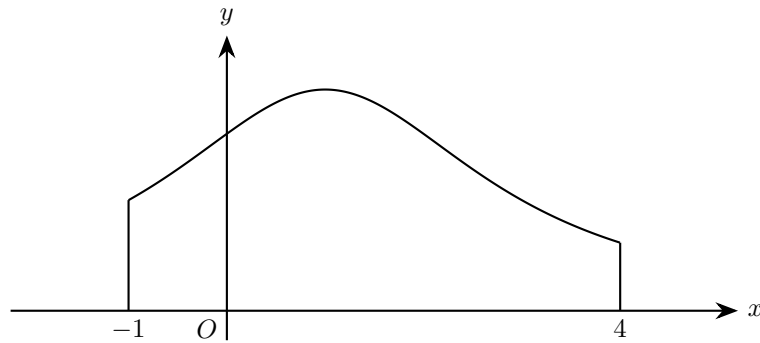
$$\int \frac{1}{(x+2)^2 + 9} dx = \frac{1}{3} \arctan\left(\frac{x+2}{3}\right)$$

Hence

$$\int \frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} dx = 2 \ln|x-1| - \frac{1}{2} \ln(x^2 + 4x + 13) + \frac{7}{3} \arctan\left(\frac{x+2}{3}\right) + c$$

So the required constants are

$$A = 2, \quad B = -\frac{1}{2}, \quad D = \frac{7}{3}$$



16. The diagram shows the cross-section of a solid wooden beam. Each unit on the axes represents 6 cm. The curved upper boundary of the beam is modelled by the equation

$$y = \frac{9}{(x-1)^2 + 4}$$

The cross-section is bounded by the curve, the x -axis and the lines $x = -1$ and $x = 4$. Assuming that the beam is a prism of length 240 cm, find, correct to 3 significant figures, the volume of the beam.

[6]

Solution

The cross-sectional area is the area under the curve

$$y = \frac{9}{(x-1)^2 + 4}$$

from $x = -1$ to $x = 4$.

First find this area in graph units:

$$A_{\text{units}} = \int_{-1}^4 \frac{9}{(x-1)^2 + 4} \, dx$$

Using the standard result

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 2$ and x replaced by $x - 1$,

$$\begin{aligned} A_{\text{units}} &= 9 \left[\frac{1}{2} \arctan\left(\frac{x-1}{2}\right) \right]_{-1}^4 \\ &= \frac{9}{2} \left[\arctan\left(\frac{x-1}{2}\right) \right]_{-1}^4 \\ &= \frac{9}{2} \left(\arctan\left(\frac{3}{2}\right) - \arctan(-1) \right) \\ &= \frac{9}{2} \left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4} \right) \end{aligned}$$

Now convert to cm^2 .

Each unit on each axis represents 6 cm, so

$$1 \text{ square unit} = 6 \times 6 = 36 \text{ cm}^2$$

Hence the cross-sectional area is

$$\begin{aligned} A &= 36A_{\text{units}} \\ &= 36 \cdot \frac{9}{2} \left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4} \right) \\ &= 162 \left(\arctan\left(\frac{3}{2}\right) + \frac{\pi}{4} \right) \\ &\approx 286.447 \text{ cm}^2 \end{aligned}$$

The beam is a prism of length 240 cm, so

$$\begin{aligned} V &= 240A \\ &= 240 \times 286.447 \\ &\approx 68747.3 \text{ cm}^3 \end{aligned}$$

Therefore, correct to 3 significant figures, the volume of the beam is

$$6.87 \times 10^4 \text{ cm}^3$$

17. Find

$$\int_1^e \frac{1}{x\sqrt{4 - (\ln x)^2}} dx$$

expressing your answer in terms of π .

[8]

Solution

Using the standard result

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin\left(\frac{x}{a}\right) + c$$

with $a = 2$ and x replaced by $\ln x$, noting that $\frac{d}{dx}(\ln x) = \frac{1}{x}$,

$$\int_1^e \frac{1}{x\sqrt{4 - (\ln x)^2}} dx = \left[\arcsin\left(\frac{\ln x}{2}\right) \right]_1^e$$

Therefore

$$\left[\arcsin\left(\frac{\ln x}{2}\right) \right]_1^e = \arcsin\left(\frac{1}{2}\right) - \arcsin(0) = \frac{\pi}{6} - 0 = \frac{\pi}{6}$$

Hence the value of the integral is

$$\frac{\pi}{6}$$